1. Write the recurrence describing the following algorithm's runtime.

```
Alg(array A of length n):
  If n = 1:
      Return A
  Else:
      Return Alg(array formed by taking the values at odd indices of A)
```

Solution:

$$T(n) = \begin{cases} 1 & \text{if } n = 1 \\ T(\frac{n}{2}) + n & \text{if } n > 1 \end{cases}$$

Note: the n non-recursive work accounts for the runtime to find the odd indices of A. We also accepted 1 as the non-recursive work, in case you considered it constant time to process the array.

■

2. Suppose you are given a stack of n pancakes of different sizes. You want to sort the pancakes so that smaller pancakes are on top of larger pancakes. The only operation you can perform is a *flip*—insert a spatula under the top k pancakes, for some integer k between 1 and n, and flip them all over.

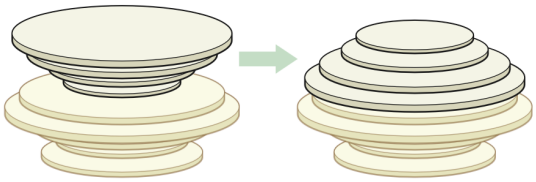


Figure 1. Flipping the top four pancakes.

(a) Let $PancakeSort$ be an algorithm that sorts a stack of pancakes. Given a stack of n pancakes S, describe how you could perform a constant number of flips on S so that calling $PancakeSort$ on the top $n-1$ pancakes completes the sorting of S.

Solution: We want to get the largest pancake to the bottom of the stack. To do so, we can find the largest pancake, insert our spatula just below it, and flip it to the top, and then flip the entire stack so that it is on the bottom. Now, we can run $PancakeSort$ on the top $n-1$ pancakes to complete the sorting of S, as required. ■

(b) Write the full recursive $PancakeSort$ algorithm.

Solution:

```
Alg(stack S of n pancakes):
  If n = 1:
      Return S
  Else:
      Find the largest pancake in S and flip it to the top of the stack.
      Flip the entire stack so that the largest pancake is on the bottom.
      Return Alg(stack formed by taking the top n − 1 pancakes of S)
```

■

(c) Write the recurrence for your $PancakeSort$ algorithm's runtime $T(n)$.

Solution:

$$T(n) = \begin{cases} 1 & \text{if } n = 1 \\ T(n-1) + n & \text{if } n > 1 \end{cases}$$

Note: the n non-recursive work accounts for the runtime to find the largest pancake and perform the necessary flips. ■